Pokhara University

Everest Engineering College

Final Internal Assessment
Fall-2025

Level: Bachelor **F.M. 100**

Program: BE CMP(5th Sem) **P.M. 45**

Section: **Time: 3hrs**

Subject: Software Engineering

Attempt all the questions.

1	a)	Define software Engineering. How do you identify risk in software engineering? Explain risk mitigation, monitoring and management.	8
	b)	Compute the function point, productivity, documentation, and cost per function for the following data: No. of user input = 24 No. of user output = 46 No. of user inquiries = 8 No. of files = 4 No. of external interfaces = 2 Effort = 36.9 P-M (person Month) Technical document = 265 pages User document = 122 pages Cost = $ 7744 per month Various processing complexity factors are: 4, 1, 0, 3, 3, 5, 4, 4, 3, 3, 2, 2, 4, 5	7
2	a)	What do you mean by agile development? Suppose Pokhara University is planning to make a software application for managing programs, notices and extracurricular information. This application needs to be completed in a span of 6-9 months for information sharing only and the rest of the features will be added in around 2 years. Propose a software development model for this software and justify your answer.	7
	b)	What are the types of software requirement? Explain requirement engineering process in detail.	8
3	a)	Draw use case and DFD level 0 and level 1for online banking system	8
	b)	What are the design modeling principles? Explain any two Architectural design styles.	7

Available on notes.arpanadhikari7.com.np

4 a) Differentiate between OOA and OOD. Explain security by design. 8

b) What is software testing? Explain different types of development testing in detail. 7

5 a) Write down four steps for basis path testing? Draw Control flow graph and calculate the cyclomatic complexity of the following function. 7

```
While(x<y){
If (x==3)then
printf (" ..");
  else
printf ("..");
}
X++;
```

8

b) Define software reliability. Explain the activities of SQA (software quality assurance) group for achieving high quality end product. 7

6 a) What is SCI (software configuration item)? Explain configuration management activities in details. 8

b) Why software reuse is important ? Explain need and benefits of using Artificial Intelligence(AI) in software engineering. 7

7 Write short notes on: (Any two) 2*5=10

a) Scrum

b) Design patterns

c) Software maintenance and its types.

notes.arpanadhikari7.com.np

Good Luck

Scanned with CamScanner

Available on notes.arpanadhikari7.com.np

COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test I

Date: 2082/09/28			
Level	BE	Full Marks	50
Programme	BCE	Time	
Semester	V	1.5 hrs	

Subject: - Software Engineering

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ *Attempt All questions.*
- ✓ *The figures in the margin indicate **Full Marks**.*
- ✓ *Assume suitable data if necessary.*

1. a) Describe Software Project Management (SPM). 3
   b) Explain the Four P's of Software Project Management in detail. 5

2. a) What is Line of Code (LOC)? Describe its demerits. 3
   b) A web-based News Portal system is developed with the following details:
   - Number of External Inputs (EI) = 30
   - Number of External Outputs (EO) = 20
   - Number of External Inquiries (EQ) = 15
   - Internal Logical Files (ILF) = 8
   - External Interface Files (EIF) = 5

   The project is assumed to have average complexity. For average complexity,

Component Type	Average
External Input (EI)	4
External Output (EO)	5
External Inquiry (EQ)	4
Internal Logical File (ILF)	10
External Interface File (EIF)	7

   If the productivity of the development team is 25 Function Points per Person-Month and the salary of a developer is Rs. 18,000 per month, calculate the total Function Points and estimate the total development cost of the system. 7

3. What are the five values of Extreme Programming (XP) in Agile? Explain Extreme Programming (XP) in detail with a necessary diagram. 8

notes.arpanadhikari7.com.np

4. Under what conditions would you choose the Rapid Application Development (RAD) process model? Explain the RAD model in detail. 8

5. a) What is Data Modeling? 2
   b) Explain the Use Case Diagram with a suitable example. 6

   OR

   Draw the Data Flow Diagram (DFD) for an Online Food Ordering System:
   i. Draw the Level-0 DFD (Context Diagram). 4
   ii. Draw the Level-1 DFD showing major processes, data stores, and data flows. 4

6. Write short notes on (any two) of the following [2*4]
   i) Software Risks
   ii) Waterfall Model
   iii) Requirements Engineering Process

Scanned with CamScanner

Available on notes.arpanadhikari7.com.np

COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test II

Exam	BE	Full Marks	40
Programme	BCE	Time	
Semester	V	1.1 hrs	

Subject: Software Engineering

- Candidates are required to give their answers in their own words as far as practicable
- Attempt ALL questions.
- The figures in the margin indicate Full Marks.
- Assume suitable data if necessary

1. What is architectural design in software engineering? Explain the concept of layered architecture in detail with suitable example and diagram. [8]

2. Define software quality. Explain the Capability Maturity Model Integration (CMMI) framework and describe its five maturity levels. [8]

OR

Explain interface design, component design, and database design in detail. How do they contribute to the development of high-quality software systems? [8]

3. Discuss the process of change control in software configuration management. Explain how it helps in maintaining the integrity of software throughout its lifecycle. [8]

4. Explain the importance of validation testing in software development. Draw the control flow graph and calculate the cyclomatic complexity of the following function [8]

```
int fun(int x, int y) {
while (x != y) {
 if (x > y) {
    x = x - y;
    print "x is greater than y";
 } else {
    y = y - x;
    print "y is greater than x";
 }
 }
 }
```

OR

Explain the terms verification and validation. Justify why Alpha testing is referred to as "developer-site testing" and Beta testing as "field testing" with suitable examples.

5. Define software reuse. Describe the different approaches to software reuse with appropriate examples. [8]

6. Write **short notes** on **(any two)** of the following [2*5=10]
   i) Object-oriented system.
   ii) **Artificial Intelligence (AI)** in Software Engineering
   iii) **Client-Server Architecture**

notes.arpanadhikari7.com.np

Scanned with CamScanner

Available on notes.arpanadhikari7.com.np

GANDAKI COLLEGE OF ENGINEERING & SCIENCE

Semester Fall

Level: Bachelor
Programme: BE CE V
Course: **Software Engineering**

Year : 2025
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1.	a)	A project size of 250 KLOC is to be developed. Software development team has average experience on similar type of projects. The project schedule is not very tight. Calculate the effort, development time, average staff size and productivity of the project.	8
	b)	How can Artificial Intelligence (AI) be integrated into software engineering practices to improve development efficiency and quality?	7
2.	a)	What is Software Process Model? Explain about iterative process model along with its advantage.	8
	b)	How does the Agile software development methodology accommodate the challenges of scaling? Explain.	7
3.	a)	What is Requirement Engineering? Explain Requirements Modeling Principle.	8
	b)	Explain Context Modeling and Behavioral Modeling in Requirement Engineering.	7
4.	a)	Explain the concept of client-server architecture, describing its components, how it works, and its benefits, with an example.	7

OR

notes.arpanadhikari7.com.np

Scanned with CamScanner

Available on notes.arpanadhikari7.com.np

Explain the concept of Layered Architecture Design. Describe its structure with a neat diagram and discuss its advantages in software system development.

b) What is Design Pattern? Explain different type of design patterns. 8

5. a) What is Software Testing? Differentiate Software Validation and Software Verification? 8

b) What is test case? Calculate cyclomatic complexity for the given code. 7

```
IF A = 354
THEN IF B > C
THEN A = B
ELSE A = C
END IF
END IF
PRINT A
```

6 a) What is Capability Maturity Model Integration (CMMI), and how does it improve process efficiency in organizations? 7

OR

What are Software Reviews? Explain Software Inspections Process.

notes.arpanadhikari7.com.np

b) What is a Version Management System? and how does it help in tracking and managing changes in software development projects? 8

7. Write short notes (on ***any two)*** 2*5

i) COTS Product Reuse

ii) Scenario Testing

iii) Software Engineering Ethics

Scanned with CamScanner

Available on notes.arpanadhikari7.com.np

Pokhara Engineering College
Internal Assessment

Level	: Bachelor	**Year:** 2026
Program	: Computer (Sem 5)	**Full Marks:** 100
Subject	: Software Engineering	**Pass Marks:** 45
		Time: 3 hrs.

Candidates are required to give answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Software and software risks. Explain the 4 Ps in Software Project Management. 7

   b) What is software project cost estimation? Differentiate between FP based estimation and COCOMO estimation. 8

2. a) Describe Waterfall model with its strength and weakness. 7

   Or

   Define agility. How SCRUM model can be used for software development. Explain.

   b) What is Software Review? Explain the different types of Software Reviews in detail. 8

3. a) What are functional and non-functional requirements? How Software Requirement Specification (SRS) is prepared? 7

   b) Draw a use case diagram for below scenario. 8

   A customer visits online shopping portal. A customer may buy item or just visit the page and logout. The customer can select a segment, then a category and brand to get different products in the desired brand. The customer can select product for purchasing. The process can be repeated for more items. Once the customer finishes selecting the product/s, the cart can be viewed. If the customer wants to edit the final cart it can be done here. For final payment, the customer has to login to portal. If the customer is visiting for the first time, he must register with the site, else the

notes.arpanadhikari7.com.np

Scanned with CamScanner

Available on notes.arpanadhikari7.com.np

customer use login page to proceed. Final cart is submitted for payment and card details and address details are to be confirmed with customer. Customer is confirmed with the shipment id and delivery of goods within 15 days.

4. a) How class diagram is constructed? Explain the entire symbols, attributes, operations and relationships with reference to a Library Management System. 7

Or

Discuss on a sequence diagram for a ATM system.

b) What are the Design Modeling principles? Explain the Layered Architecture with suitable diagram. 8

5. a) Differentiate between White Box and Black Box testing. 7

b) What is software validation and software verification? Explain the Boundary Value Analysis (BVA) mechanism, with a code example and design a test case for it. 8

6. a) What are software configuration items (SCI)? Explain the Version control in Software Configuration Management (SCM) process. 7

b) Introduce Software reuse. Discuss the advanced software engineering trends in brief. 8

7. Write short notes: (any two) 10

notes.arpanadhikari7.com.np

a) Object Oriented Design

b) Extreme Programming

c) Cyclomatic Complexity

Scanned with CamScanner

Available on notes.arpanadhikari7.com.np

POKHARA UNIVERSITY
FACULTY OF SCIENCE & TECHNOLOGY
SCHOOL OF ENGINEERING

Exam	Final Internal Examination 2025 Fall		
Level	B.E.	F M	100
Program	Computer	PM	45
Year/ Part	III/I	Time	3 Hrs

Subject: Software Engineering

Candidates are required to give answers in their own words as far as practicable.
The figure in the margin indicates full marks. Assume suitable data if necessary.
Attempt all the questions.

1. a) Explain the nature and characteristics of software. How are they different from traditional engineering products? 7
   b) If you were Project manager, How would you estimate the cost for your project? Explain by choosing one of the estimation techniques with numerical example and justification for choosing this estimation. 8

2. a) Draw and explain the Spiral Model highlighting its risk analysis phase. 8
   b) Why is Scrum preferred in fast-changing environments? Compare it with the Incremental Model using practical examples. 7

3. a) Explain the Requirements Engineering Process from feasibility study to Management. 7
   b) Draw a Context Diagram for an Online Library Management System and explain how it helps in requirements elicitation. 8

   OR

   Draw a Behavioural Model (eg: State Diagram) for an ATM system and explain each state.

4. a) Discuss Security by Design principles and explain how they reduce system vulnerabilities. 7
   b) Explain the Layered Architectural Style with a neat diagram. Mention its advantages. 8

   OR

   Explain the Client-Server Architecture with a neat diagram and its real-world applications.

notes.arpanadhikari7.com.np

5. a) Given the following program. Make a control flow graph. 7

```
IF A = 354
   IF B > C
      A = B
   ELSE
      A = C
PRINT A
```

   Calculate the Cyclomatic Complexity to explain the minimum number of test cases required . Also explain the importance of basis path testing. 8

Scanned with CamScanner

Available on notes.arpanadhikari7.com.np

48

b) Explain Equivalence Partitioning and Boundary Value Analysis with suitable examples.

6. a) Discuss the role of software standards such as International Organization for Standardization and CMMI Institute in improving software processes. 7

b) If change requests continue increasing rapidly, explain how poor version management could affect testing, delivery, reliability, and system releases. 8

7. Write short notes on: *(Any two)* 2×5

a) Software Reliability

b) Cloud-Based Software Engineering

c) Acceptance Testing

*** ***Best of Luck*** ***

notes.arpanadhikari7.com.np

req
integrity
Bending (alloy)
Reliabilir

Scanned with CamScanner

Available on notes.arpanadhikari7.com.np

NEPAL COLLEGE OF INFORMATION TECHNOLOGY
Assessment Fall 2025

Level: Bachelor | Year : 2025
Programme: **Computer 5th (Day/Mor.)** | Full Marks: 100
Course: **: Software Engineering** | Pass Marks: 45
Semester:**V** | Time : 3 Hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

Q.n 1.a	Define software. Explain some ethics that need to follow by software engineers.	2+5
B	How can empirical estimation model be used for software cost estimation Calculate risk exposure from following case. **Risk Identification:** Only 60 percent of the software components scheduled for reuse will, in fact, be integrated into the application. The remaining functionality will have to be custom developed. **Risk Probability:** 70% Risk Impact: 80 reusable software components were planned. The average component is 350 LOC and the software engineering cost for each LOC is RS. 20.0	8
Q.n 2.a	Which software models you prefer when software need to deliver within 2 to 3 months and why. Explain	7
b	Define functional and non-functional requirements. Explain requirement engineering process. ?	8
Q.n 3.a	NCIT College wants to develop Student Management System. The system will allow: • Students to login • Students to view academic calendars • Students to view and submit assignments • College admin to add, remove students and teachers. • Teachers to upload tutorial and assignment The system also stores: • Student details • Teachers details	15

Available on notes.arpanadhikari7.com.np

From above scenario draw:

a. **Use case diagram**

b. **Class diagram**

c. **Sequence diagram**

Q.n 4 Write down steps for basis path. Calculate the Cylomatic Complexity of the following code snippet: 8

```
while(condition)
{
  if(condition)
  {
    if body
  }
  else if (condition)
  {
    else if body
  }
 value update
}
```

b Define software testing. What is integration testing why integration testing is important explain 7

Q.n 5.a Define software reliability. Explain three phases involved in the software quality reviews. 8

b Define software configuration management. Explain change management process in detail. 7

Q.n 6.a What are the fundamental sources of change in any software project? Define software maintenance and explain different types of software maintenance with suitable example 7

notes.arpanadhikari7.com.np

b What is reuse based software engineering? What are the benefits and problems associated with software reuse? 8

or

What is cloud based software engineering? Explain how can cloud services be used in software engineering?

Q.7 Write short notes on any of two[2*5 = 10]

a. Software frameworks

b. Security by design

c. Agile principles

Scanned with CamScanner

Available on notes.arpanadhikari7.com.np

NEPAL ENGINEERING COLLEGE

Level: Bachelor | Year : 2025
Programme: BE | Full Marks: 100
Course: Software Engineering | Pass Marks: 45

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. (a) Why do we need to follow software process during software development? If customer changes requirements frequently which software process do you follow and why. 8

   (b) Discuss the benefits of cost and time estimation during project planning. 7

   A university administration system is to be developed, and the estimated lines of code (LOC) are calculated to be 90,000. Based on historical data for educational domain projects, the average productivity is 400 LOC per person-month (PM), and the labor rate is Rs. 30,000 per month. Calculate the estimated effort and estimated project cost for the development of the university administration system.

   OR

   Explain the demerits of Lines of Code (LOC).

   Given the data below, compute the function point value, effort and total cost of a project with the following information domain characteristics.

   Number of user inputs: 35
   Number of user outputs: 55
   Number of user inquiries: 5
   Number of files: 4
   Number of external interfaces: 3

   Assuming that the complexity of given software is simple, productivity of software developers is 20 FP/PM and their salary is Rs.200/PM.

2. (a) How does use case help to understand system requirements? Draw use case diagram for online food delivery system. 7

   (b) What is SS (Software Requirement Specification) document Explain components of SRS document. 8

3. (a) Write down five design principles. Explain how coupling and cohesion helps in effective modular design. 8

   (b) What is architecture design? What are the different types of architectural style? 7

Available on notes.arpanadhikari7.com.np

4. (a) Draw the control flow graph and calculate the cyclomatic complexity of the following function. 7

```
int a 1, b-1, n, c;
int i=1;
while (i < n) {
c = a+b;
a = b;
b = c;
printf("%d", c);
i++;
}
```

(b) What is black box testing? Explain Equivalence Partitioning with an example. 8

5. (a) What is FTR and SQA? What are the objectives of FTR? 8

(b) What are the fundamental sources of changes in any software system? Explain how version control process supports software configuration management. 7

6. (a) Define object-oriented paradigm. Explain how do you identify the elements of an object model. 7

(b) Explain object design process in brief. 8

OR

7. Explain system design process in brief. 2*5

Write Short Notes on any two

a) 4Ps of Management Spectrum
b) Types of risk
c) Capability maturity model

notes.arpanadhikari7.com.np

CamScanner

Scanned with CamScanner

Available on notes.arpanadhikari7.com.np

UNITED TECHNICAL COLLEGE

Level: Bachelor | Year: 2025

Program: Computer | Full Marks: 50

Course: Software Engineering | Pass Marks: 25

Time: 1.5 Hrs.

Attempt all the questions.

1. [7+8=15]

   a. What is risk in a software? How do you identify risk in software engineering? Explain risk mitigation, monitoring and management.

   b. A technology company is developing an advanced AI-powered healthcare system. The project is highly complex, involving multiple uncertainties, evolving requirements, and significant risks. It will take approximately three years to complete, with a team of 50 developers, data scientists, and domain experts. The company seeks a software engineering model that allows iterative development, risk assessment, and continuous refinement. Propose a suitable model and justify your choice.

2. [7+8=15]

   a. Define outsourcing. Suppose a software project has an estimated 58,600 lines of code (LOC). The organization's average productivity is 620 LOC per person-month (PM), and the labor cost per developer per month is Rs 12,000. Calculate the estimated effort required to complete the project and determine the total project cost.

   b. Define Requirement. Explain the requirement engineering process in detail.

3. [8+7=15]

   a. A patient goes to registration machine then he presses on button and then screen opens. After that he enters the patient id number and books for the doctor for his checkup. After that he checks the category of disease from given list then chooses doctor's name from given doctor's name list, then he enters time he wants to meet with

Scanned with CamScanner

Available on notes.arpanadhikari7.com.np

doctor. For this registration he needs to enter the amount, if the amount is ok it accepts the registration and prints the registration slip otherwise it gives an alarm.

From above information

1. Draw a use case diagram
2. Draw a sequence diagram

b. Explain the concept of the Class Diagram along with an example of a class diagram for the Online Shopping System.

OR

Draw an activity diagram to borrow books from your respected college library management system.

4. [8+7=15]

a. Define the architectural design. Discuss the layered architecture model with the help of an example.

b. Explain the application of artificial intelligence in software engineering.

5. [7+8=15]

a. Compare and contrast verification with validation. Explain how the unit testing is carried out.

b. Explain basic path testing. Compute Cyclomatic Complexity from the given piece of program.

```
int fun(int x, int y)
{
 while(x != y)
{
  if(x > y)
   x = x-y;
 else y = y-x;
}
```

Scanned with CamScanner

Available on notes.arpanadhikari7.com.np

return

6. [7+8=15]

   a. Discuss the SPI framework. Explain CMMI in detail.

   b. What is configuration management? Explain how Version Management keeps track of all the different versions of software components (like files, code, or documents).

7. Write short notes on: (Any four)
   [4*2.5=10]

   a. ISO

   b. Reviews and Inspections

   c. Cloud-Based Software Engineering

   d. Individual Vs integrated application systems

   e. Inversion of Control (IoC)

   f. COCOMO

notes.arpanadhikari7.com.np

Scanned with CamScanner

Available on notes.arpanadhikari7.com.np

Universal Engineering and Science College
Chakupat, Lalitpur
Pre-Board Examination

Level: Bachelor
Programme: B.E
Course: Software Engineering
Semester: V

Year: 2026
Full Marks: 100
Pass Marks: 45
Time: 3hrs

Attempt all the questions

1 a. Define software and it characteristics. What are software myths? Explain different types of software myths. 7

b. Explain different types of software risk? .Suppose an estimated LOC on any software project is 58,000, organizational average productivity is 620 LOC/PM. And the labor rate per month is Rs 12,000. Calculate the total project cost and estimated effort. 8

Or

Compute the function point value for a project with the following information.
Number of inputs: 50
Number of outputs: 40
Number of inquiries: 35
Number of files: 06
Number of external interfaces: 04
Assume that all complexity adjustment values are average. If the productivity of the company is 6.5 FP/PM and average salary is $800 per month then calculate the cost and effort required for the project. Assume necessary data if required.

2 a. What are the reasons behind agile popularity? Which software development model is best suited or risk-driven software development? 8

b. Draw a level 0 and level 1 DFD for online learning system (OLS). 7

3 1. Differentiate between include and extend relationship. Draw a use case diagram from the given case study. 8

In hospital a patient goes to registration machine he press the ON button then the screen open. He enters patient ID number. He books for the doctor for checkup. He checks the category of disease from given list, then he chooses the doctor's name from given doctor's name list, he enters time he wants to meet with doctor. For this registration he need to enter the amount, if the amount digit is ok it accepts the registration and prints the registration slip, otherwise it will give a signal of alarm insufficient amount.

or

Differentiate between aggregation and composition. Create a class diagram for online shopping system

Available on notes.arpanadhikari7.com.np

a. List out and explain various architectural design with suitable diagrams. 7

4 a. Describe design modeling principles. Differentiate between OOA and OOD. 8

b. Why testing is necessary? What are different considerations during unit testing of any module? Explain importance of drivers and stub while performing the unit testing? 7

5 a. Explain the Boundary Value Analysis (BVA) mechanism for software testing with suitable example and also design a test case for it. 8

b. What is ISO standard? How can a project manager assure a client that the software product is of high quality? Describe the elements of SQA. 7

Or

How can you measure software reliability and availability? What are the set of guidelines for Formal Technical Review (FTR)?

6

a. Why SCM is essential during software development? Discuss the SCM process. 7

b. What are the benefits of Software reuse? Discuss the role of cloud computing and AI in software Engineering. 8

or

Define auditing and reporting. What are the reasons for SPI (Software Process Improvement)? How SPI is performed?

7 Write short notes on (Any Two) 2*5

a. Waterfall model
b. Alpha Vs Beta Testing
c. RMMM and RMMM plan
d. People CMM

notes.arpanadhikari7.com.np

Scanned with CamScanner

Available on notes.arpanadhikari7.com.np

Madan Bhandari College of Engineering

Urlabari-3, Morang

Final Internal Examination

Level: Bachelor — Full Marks: 100

Programme: B.E(Computer) — Pass Marks: 45

Year/Part: III/I — Time: 3 hrs

Subject: - Software Engineering(Set-A)

- ✓ *Candidates are required to give their answers in their own words as far as possible.*
- ✓ *Attempt all question*

1. a) Define Software and it characteristics. What are software myths? Categorize and explain different types of software myths with suitable examples. 7

   a) Why software engineers need to follow ethics? Discuss in brief Draw the control flow graph and calculate the cyclomatic complexity of the given piece of code. 8

```
large=x[0];
        for(i=1;i<=n;i++)
        {
          if(x[i]>large)
            large=x[i];
        }
```

2. a) Compare the **Incremental** and **Iterative** software process models with examples. 7

OR

Define agility. How SCRUM model can be used for software development. Explain.

b) Explain the significance of **reviews and inspections** in the software quality assurance process. 8

3.a) How class diagram is constructed? Explain the entire symbols, attributes, operations and relationships with reference to a Library Management System. 8

OR

Draw a sequence diagram for below ticket booking process. A user searches for available flights by entering travel details. The system contacts the airline database and shows available flights. The user selects a flight and provides passenger details. The system requests payment from the payment gateway. Upon successful payment, the ticket is issued, and a confirmation email is sent.

b) Explain the Layered Architecture and Repository architecture with suitable diagrams. 7

4. a) Differentiate between verification and validation. Explain the different levels of software testing in brief. 7

b) Compute the function point value for a project with the following information: 8

- Number of inputs: 44
- Number of outputs 80
- Number of inquiries: 30
- Number of files: 20
- Number of external interfaces: 4

Assume that all complexity adjustment values are average. A review reveals that the average productivity is 40 FP/PM and the labor rate is Rs. 27440 per month. Calculate estimated project cost and estimated effort for the software.

Available on notes.arpanadhikari7.com.np

5 a) Define Software Configuration Management (SCM). Explain its activities in detail. 7

b) Introduce Software reuse. Discuss the role of Cloud computing and AI in Software Engineering. 8

6. a) Obtain Context and 1 -level DFD for the Food Ordering System 8

Customer can place an Order of the food. The Order Food process receives the Order. forwards it to the Kitchen, store it in the Order data store, and store the updated Inventory details in the Inventory data store. The process also delivers a Bill to the Customer. Manager can receive Reports through the Generate Reports process, which takes Inventory details and Orders as input from the Inventory and Order data store respectively Manager can also initiate the Order Inventory process by providing Inventory order. The process forwards the Inventory order to the Supplier and stores the updated Inventory details in the Inventory data store.

b)Explain the Boundary Value Analysis (BVA) mechanism for software testing with a suitable example and, also, design a test case for it. 7

7.Write short notes on (Any Two) 2×5

a) Functional Vs. Non-functional requirements

b) 4P's of Management spectrum

c) Version Control

notes.arpanadhikari7.com.np

Scanned with CamScanner

Available on notes.arpanadhikari7.com.np

Lumbini Engineering College
(Final Internal Exam)

Level: Bachelor
Program: BE Computer 5th
Course: Software Engineering

Year : 2026
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answer in their own words as far as practicable.
The figure in the margin indicates full marks.
Attempt all the questions.

1. a) Describe V process model with its advantages and disadvantages 7
   b) What do you mean by agile development? Why is it important? Explain the scrum software development in detail. 8

2. a) What is Software scope? Effective software project management focuses on 4P's. Explain all. 8
   b) Explain different feasibility associated with Software. . 7

3. a) What is requirement analysis? Explain different elements of requirement analysis model. 7
   b) What do you mean by scenario based modelling and behavioural based modelling? Explain with example. 8

4. a) Obtain 1-level DFD for Biometric Attendance system. 7
   b) What do you mean by design model? What are the elements of effective interface design? Explain the evaluation of the user interface with evaluation cycle diagram. 8

5. a) Demonstrate use of scenario- based testing with suitable example. 7
   b) What is Formal Technical Review (FTR)?Why is it important in SQA activities? explain . 8

6. a) What are the relationship between mean-time-between-failure(MTBF) and mean-time-to-failure(MTTF), meant-time-to-repair with service availability. 8
   b) Explain few trends, methods and tools that are likely to have influence on software engineering.

7. Write short notes on **any two:** 2×5

   a) Functional vs non-functional requirements b) People CMM c) Alpha and beta testing

Available on notes.arpanadhikari7.com.np

Madan Bhandari College of Engineering

Urlabari-3, Morang

Final Internal Examination

Level: Bachelor | Full Marks: 100

Programme: B.E(Computer) | Pass Marks: 45

Year/Part: III/I | Time: 3 hrs

Subject: - Software Engineering(Set-B)

- ✓ *Candidates are required to give their answers in their own words as far as possible.*
- ✓ *Attempt all question*

1.a) What is the **software crisis**? Differentiate between **Software Engineering and System Engineering**.Software doesn't wear out but hardware does. Justify this statement with the help of failure curve. 7

b) Why software engineers need to follow ethics? Discuss in brief. Draw the control flow graph and calculate the cyclomatic complexity of the given piece of code. 8

```
int x;
int y;
while(x<y)
{
   if(x == 5)
    printf("Correct Choice")
}
```

2.a) What do you mean by agile development? How agile development is different than traditional software development model? Explain. 7

OR

Compare the **Incremental** and ***Iterative*** software process models with examples.

b) What is risk in a software? How do you identify risk in software engineering? Explain risk mitigation, monitoring and management. 8

3.a) How class diagram is constructed? Explain the entire symbols, attributes, operations and relationships with reference to a Library Management System. 8

OR

Draw a sequence diagram for below ticket booking process. A user searches for available flights by entering travel details. The system contacts the airline database and shows available flights. The user selects a flight and provides passenger details. The system requests payment from the payment gateway. Upon successful payment, the ticket is issued, and a confirmation email is sent.

b) What is architectural design and what are the factors that determine the architectural design decisions. Explain about repository and pipe and and filter architecture style. 7

4. a)Define software testing. What are the different considerations during unit testing of any module? Explain importance of driver and stub while performing the unit testing. 7

b) Pokhara University intends to develop a software. The project size is of 400 KLOC. Software development team has average experience on similar type of projects. The project schedule is not very tight. Using COCOCMO model, calculate the Effort, development time, average staff size, and productivity of the project. 8

Project	a	b	c	d
Organic	2.4	1.05	2.5	0.38

Scanned with CamScanner

Available on notes.arpanadhikari7.com.np

Semi-Detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.2	2.5	0.32

5. a) Define software configuration items? Explain the process of system building and also mention Some common problems during continuous system integration. 7

b) Write down the some importance of software reuse. Explain the need and benefits of using artificial intelligence(AI) in software engineering. 8

6. a) Obtain 0-level and 1-level DFD for the Food Ordering System: 7

b) Define Software Quality. Explain the activities of software quality assurance(SQA) group for achieving a high quality end product. 8

7. Write short notes on (Any Two) 2×5

a) Design Patterns

b) Waterfall model

c) Black Box vs White Box Testing

notes.arpanadhikari7.com.np

Scanned with CamScanner

Available on notes.arpanadhikari7.com.np